iDisc Automation

Dataset Clarification Questions

Prepared following Exploratory Data Analysis — March 2026

Key Findings from EDA

The following metrics were derived directly from the dataset and motivate the questions in the next section:

Metric	Value / Observation
Total translators in dataset	**~982 (after cleaning)**
Translators with ≤20 tasks	**495 (50.4%) — insufficient feedback for ML**
Average tasks per translator	**1,015**
Maximum tasks by a single translator	**65,209 (64x the average)**
Average cumulative hours per translator	**1,752 hrs**
Median cumulative hours per translator	**133 hrs (mean/median gap: ~13x)**
Translators within 0–500 hours	**620 (65.3%)**
Translators within 0–5,000 hours	**845 (89.0%)**
Average translators per sector	**366.67 (high variance across sectors)**
Same SOURCE/TARGET language records	**Present — task type unclear**

Clarification Questions

#	Question	Context / Observed Signal	Impact on Automation
A. Data Sufficiency & Cold-Start			
1	**What is the minimum number of tasks considered sufficient to build a reliable translator profile for model training?**	*50.4% of translators have ≤20 tasks. Below a threshold, there is not enough signal to learn preferences, strengths, or sector fit.*	Defines who can be included in training vs. cold-start pool.
2	**Is there a recommended strategy for handling 'cold-start' translators (those with very few historical tasks)?**	*With ~495 translators having ≤20 tasks, falling back to content-based or rule-based assignment is likely needed for a significant portion of the pool.*	Architecture decision: hybrid model vs. pure ML.

#	Question	Context / Observed Signal	Impact on Automation
3	**Are there translators with 0 recorded tasks who still appear in the dataset? How should they be interpreted?**	*The dataset includes translator IDs. It is unclear whether entries with 0 tasks represent registered-but-inactive translators or data gaps.*	Data cleaning and model inclusion criteria.
B. Outliers & Distribution Anomalies			
4	**The top translator has completed 65,209 tasks while the average is 1,015. Are these high-volume translators a special category (e.g., agencies, bots, power users)?**	*A single entity at 65x the average is a statistical outlier that could distort model training if not handled separately.*	Outlier treatment strategy and data segmentation.
5	**The mean total hours per translator is 1,752 but the median is only 133. What explains this extreme skew?**	*A mean/median gap of ~13x indicates a very long right tail. A small number of translators may be inflating the average significantly.*	Normalization approach and fairness of workload metrics.
6	**Are hours self-reported or system-tracked? How reliable is the HOURS column as a workload proxy?**	*If hours are manually entered, they may be inconsistent or biased, making them an unreliable feature for the model.*	Feature reliability for training.
	99 translator rates exceed the maximum client selling price (€50). Are these valid rates, legacy data, or errors? Assigning a translator who costs more than the client pays would be a business logic failure.		
C. Language Pair Integrity			
7	**There are records where SOURCE_LANG equals TARGET_LANG (same-language tasks). What type of task does this represent?**	*Standard translation assumes different source and target languages. Same-language tasks may be editing, proofreading, or summarisation — fundamentally different task types.*	Task type taxonomy and model scope.
9	**Are there language pairs with very few historical tasks that are still expected in future assignments?**	*Low-frequency language pairs will have sparse training data, requiring special handling or exclusion from ML-based assignment.*	Model scope and fallback logic.
	One translator covers 58 language pairs. Is this realistic, or does it indicate a data quality issue (e.g., an agency account)?		
D. Sector Coverage			
10	**The average is 366.67 translators per sector, but the chart shows high variance across sectors. Are some**	*Sectors with very few translators may not have enough data for sector-specific*	Sector-specific model feasibility.

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	sectors underserved and still expected to receive tasks?	*models, or may face real supply constraints.*	
E. Task Type & Quality Labels			
12	**What is the full taxonomy of TASK_TYPE values? Are all task types expected to be automated, or only a subset?**	*Automating all task types uniformly may be inappropriate if some require human judgment in assignment (e.g., high-stakes legal vs. routine technical).*	Model scope definition.
14	**How is the quality score defined and measured? Is it a numerical scale, a categorical rating, or a derived composite?**	*Without a consistent definition, quality scores cannot be used reliably as training labels or evaluation metrics.*	Label definition and model evaluation.
F. Temporal Validity			
15	**Are there translators who were active historically but are no longer available? How should inactive translators be flagged?**	*Including inactive translators in the assignment model would produce invalid recommendations.*	Data freshness and inference-time filtering.

Next Steps

Upon receiving answers to the above questions, the following decisions can be made:

- Define a minimum task threshold to separate 'trainable' translators from cold-start cases.
- Resolve outlier treatment for extremely high-volume translators before training.
- Clarify whether QUALITY_EVALUATION can serve as a reliable training label, or if an alternative target variable is needed.
- Establish a temporal cutoff for training data to mitigate concept drift.